Age and Medical Conditions

Gender-Stratified Analysis

HPP 10K Dataset

Ages 40-70

Female: 12,313 | Male: 11,152

Blue = Male | Red = Female

EXECUTIVE SUMMARY: GENDER-STRATIFIED ANALYSIS

This report examines whether the relationship between age and medical conditions differs between males and females.

SAMPLE SIZE

- Female: 12,313 subjects (52%)
- Male: 11,152 subjects (48%)
- Age range: 40-70 years

SIGNIFICANT ASSOCIATIONS ($p<0.05$, $n \geq 50$)

MALES:
- 26 significant conditions
- 15 increase with age
- 11 decrease with age

FEMALES:
- 32 significant conditions
- 18 increase with age
- 14 decrease with age

KEY FINDINGS

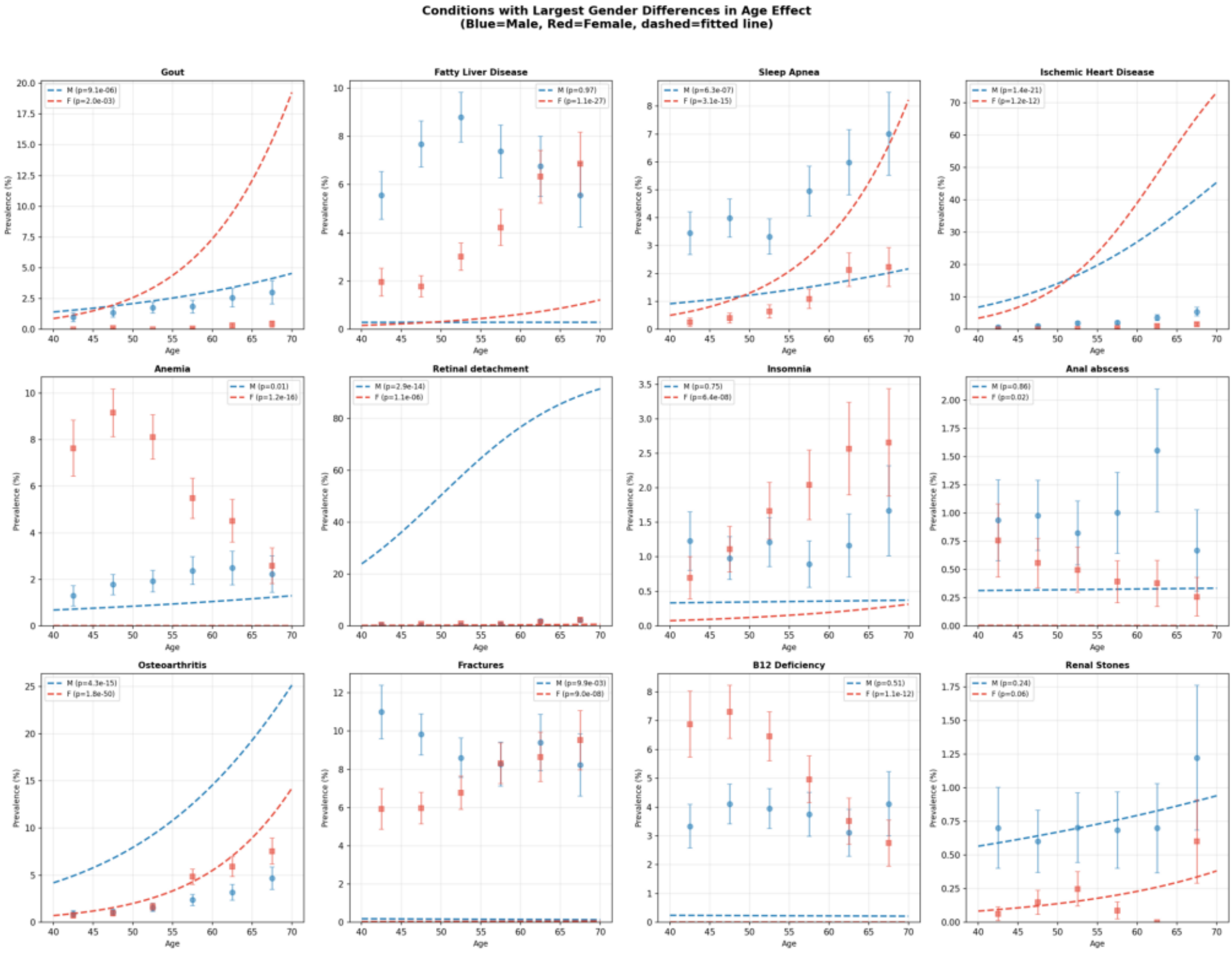
1. CARDIOVASCULAR CONDITIONS
   Generally increase with age in BOTH genders, but may have different slopes (effect sizes).

2. SEX-SPECIFIC CONDITIONS
   - Breast Cancer, PCOS, Endometriosis: Female only
   - Erectile Dysfunction: Male only
   - Gout: More common in males

3. MENTAL HEALTH
   Depression, Anxiety: May show different age patterns by gender

4. AUTOIMMUNE CONDITIONS
   Often more prevalent in females with different age trajectories

Conditions with Largest Gender Differences



Top 15 Conditions Increasing with Age - MALES

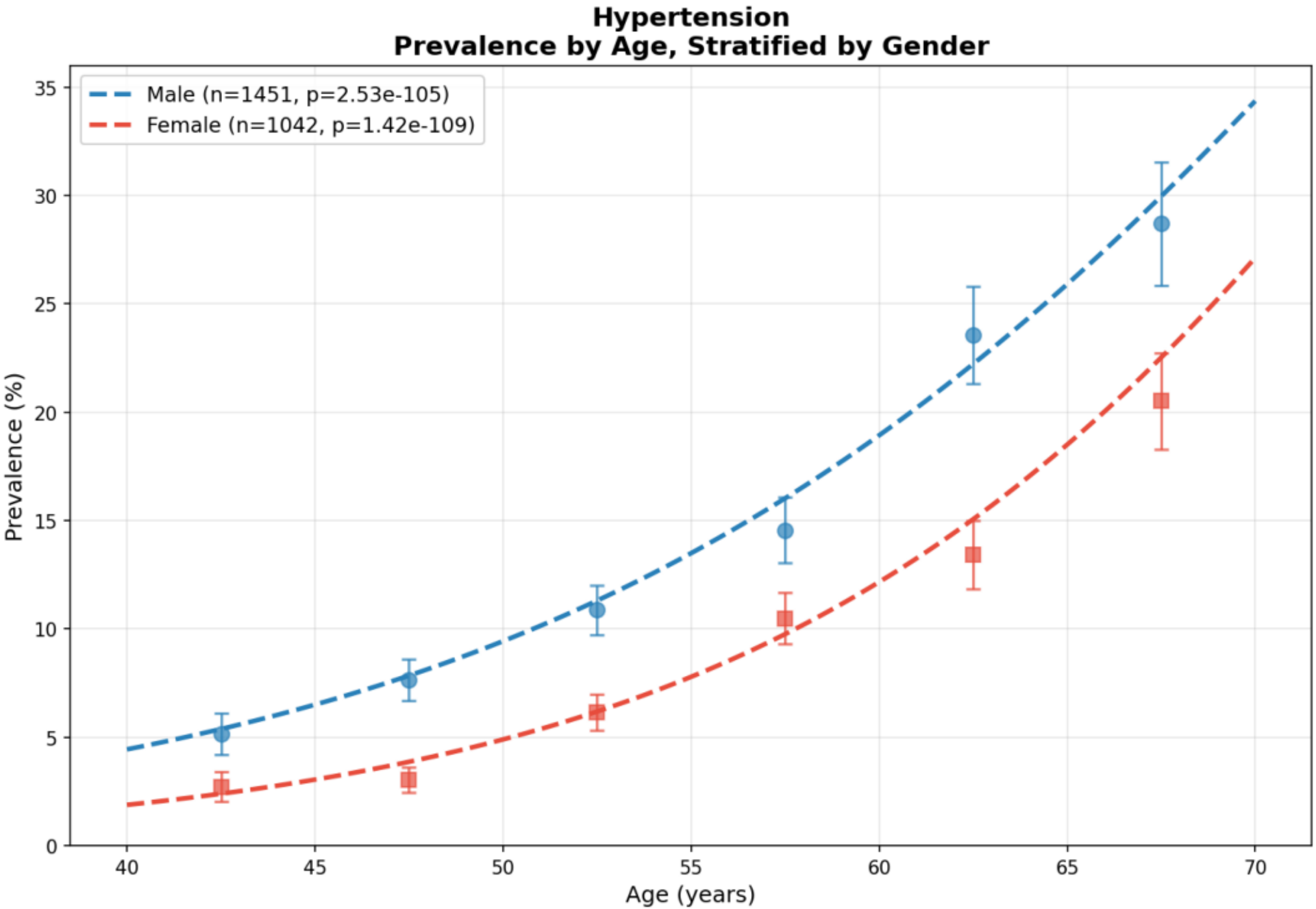
Condition	N (Male)	OR/Decade	P-value
Retinal detachment	76	3.23	2.86e-14
Ischemic Heart Disease	231	2.25	1.36e-21
Hypertension	1451	2.24	2.53e-105
Diabetes	137	2.12	4.45e-12
Glaucoma	77	2.05	6.22e-07
Osteoarthritis	216	1.97	4.34e-15
Hyperlipidemia	2751	1.79	1.39e-89
Hearing loss	457	1.72	2.02e-19
Prediabetes	903	1.71	2.42e-34
Gout	197	1.49	9.12e-06
Gallstone Disease	256	1.44	4.11e-06
Peptic Ulcer Disease	223	1.35	4.55e-04
Sleep Apnea	490	1.34	6.29e-07
Urinary Tract Stones	510	1.31	2.54e-06
Anemia	217	1.24	0.0130

Top 15 Conditions Increasing with Age - FEMALES

Condition	N (Female)	OR/Decade	P-value
Ischemic Heart Disease	54	4.25	1.23e-12
Glaucoma	84	2.87	1.71e-12
Osteoarthritis	395	2.85	1.83e-50
Hypertension	1042	2.69	1.42e-109
Sleep Apnea	123	2.61	3.10e-15
Hyperlipidemia	2261	2.15	5.07e-131
Fatty Liver Disease	456	1.97	1.14e-27
Prediabetes	1049	1.82	9.84e-46
Retinal detachment	124	1.76	1.10e-06
Hearing loss	365	1.72	1.57e-15
Diabetes	110	1.65	4.76e-05
Insomnia	211	1.62	6.44e-08
Peptic Ulcer Disease	205	1.5	7.43e-06
Hypercholesterolaemia	50	1.33	0.1185
Fractures	902	1.27	9.03e-08

Conditions with Largest Gender Differences in Age Effect

Condition	N Male	OR/10y Male	N Female	OR/10y Female
Gout	197	1.49	15	3.0
Fatty Liver Disease	813	1.0	456	1.97
Sleep Apnea	490	1.34	123	2.61
Ischemic Heart Disease	231	2.25	54	4.25
Anemia	217	1.24	830	0.66
Retinal detachment	76	3.23	124	1.76
Insomnia	125	1.04	211	1.62
Anal abscess	109	1.02	59	0.66
Osteoarthritis	216	1.97	395	2.85
Fractures	1034	0.89	902	1.27
B12 Deficiency	417	0.96	693	0.68
Renal Stones	79	1.19	22	1.66
Glaucoma	77	2.05	84	2.87
Fibromyalgia	34	0.91	211	1.18
Diabetes	137	2.12	110	1.65


Hypertension
Prevalence by Age, Stratified by Gender
Male (n=1451, p=2.53e-105)
Female (n=1042, p=1.42e-109)
Prevalence (%)
Age (years)
0
5
10
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25
30
35
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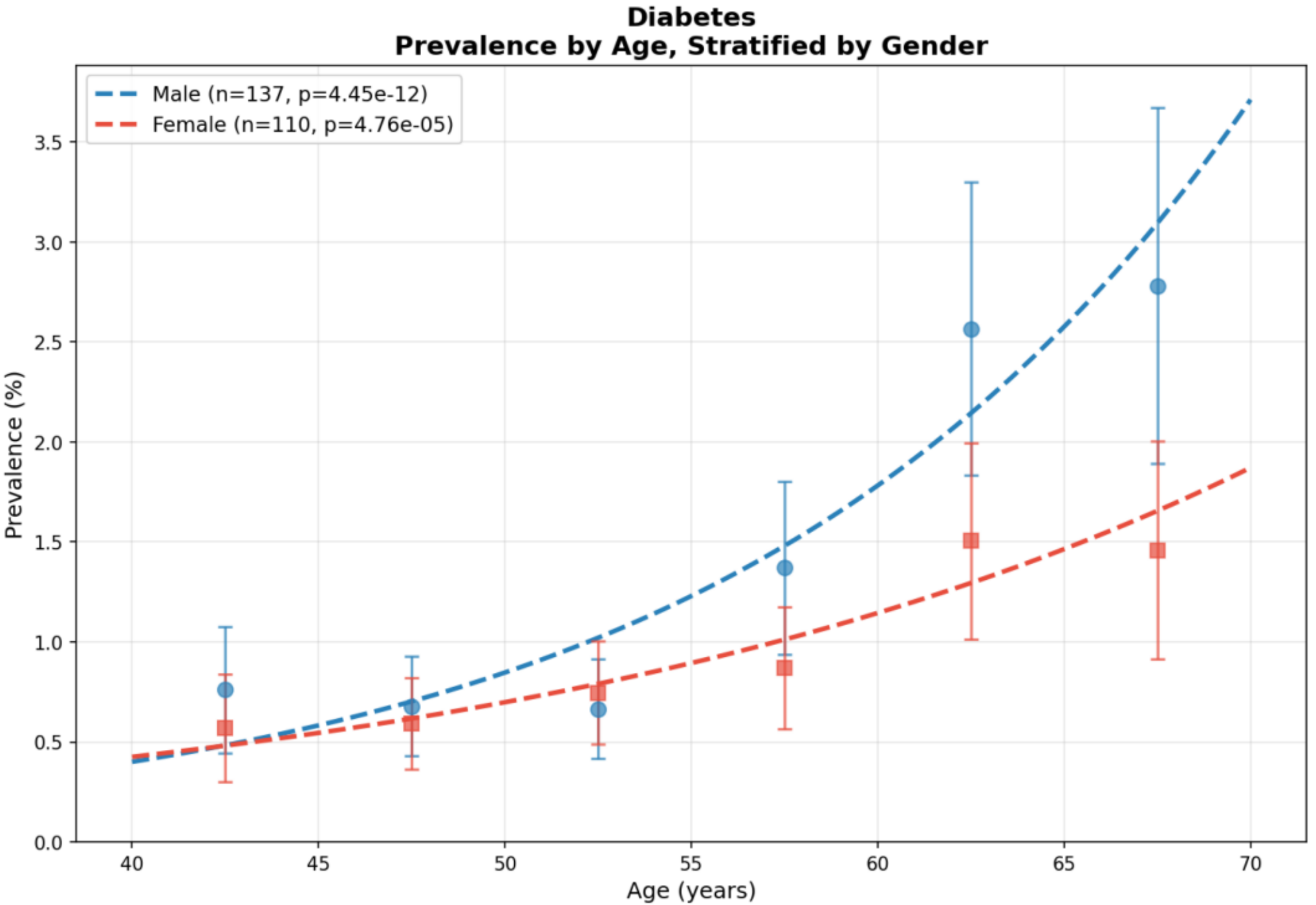
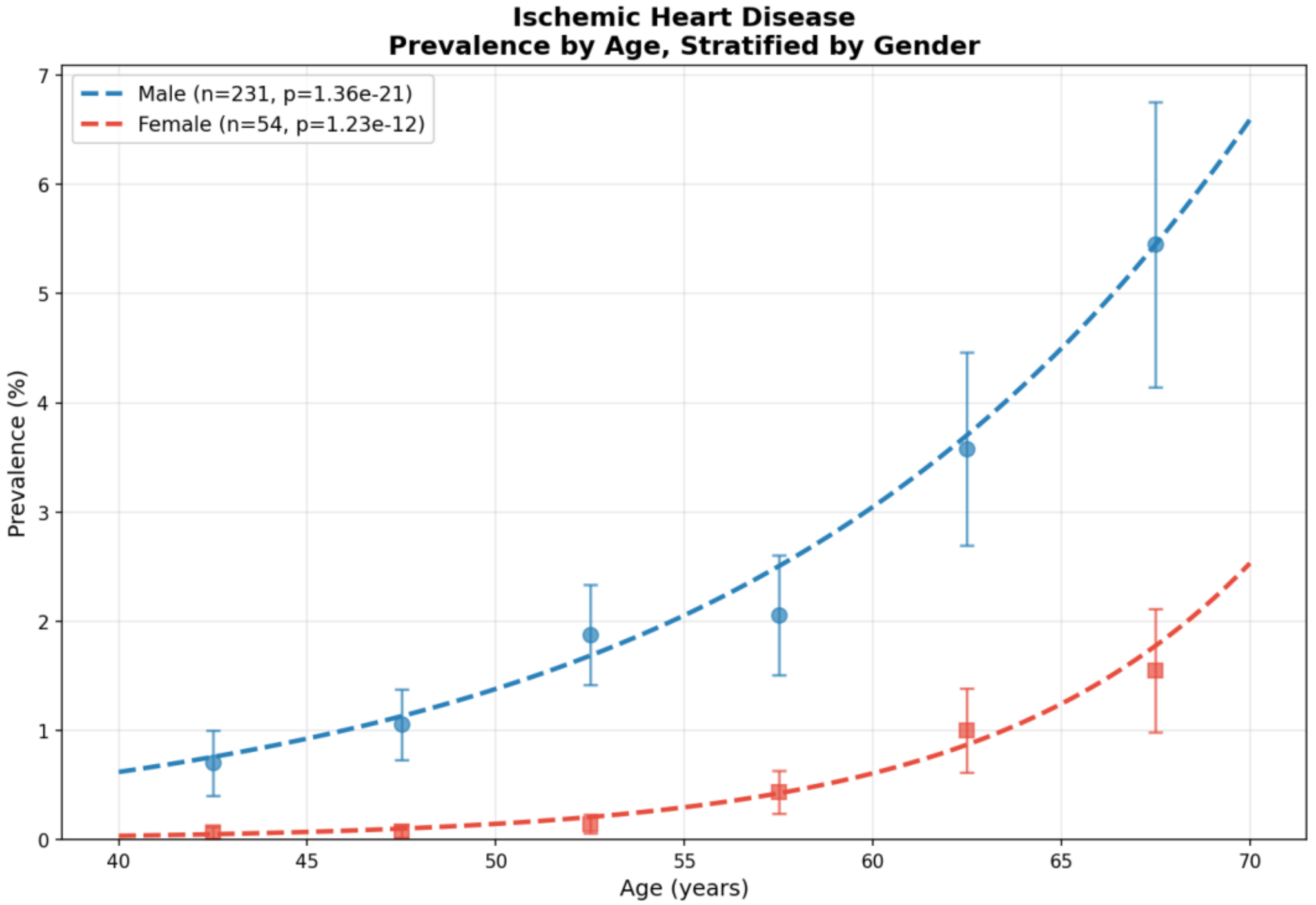
70


Diabetes
Prevalence by Age, Stratified by Gender
Male (n=137, p=4.45e-12)
Female (n=110, p=4.76e-05)
Prevalence (%)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
40
45
50
55
60
65
70
Age (years)


Ischemic Heart Disease
Prevalence by Age, Stratified by Gender
Male (n=231, p=1.36e-21)
Female (n=54, p=1.23e-12)
Prevalence (%)
Age (years)
0
1
2
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4
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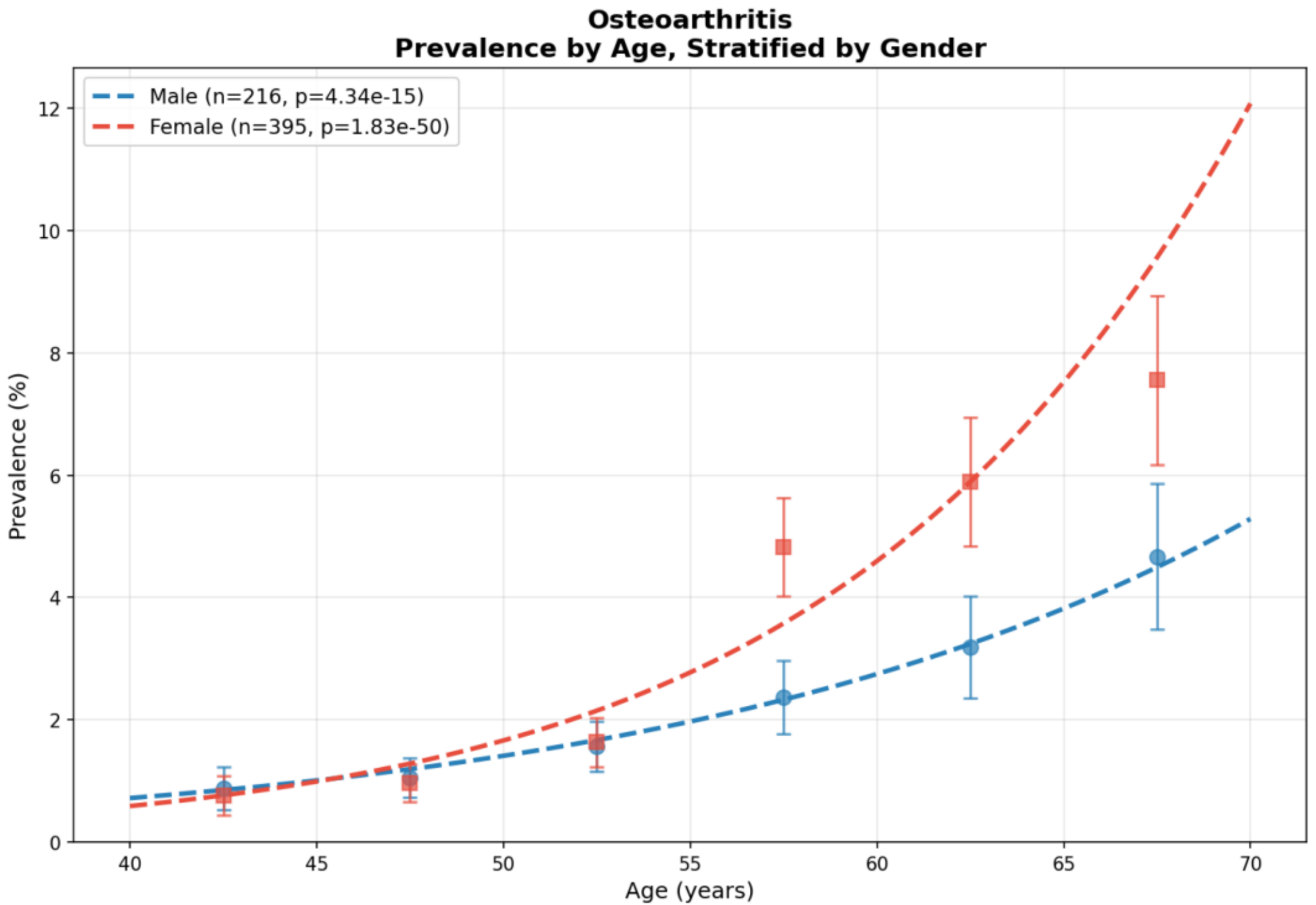
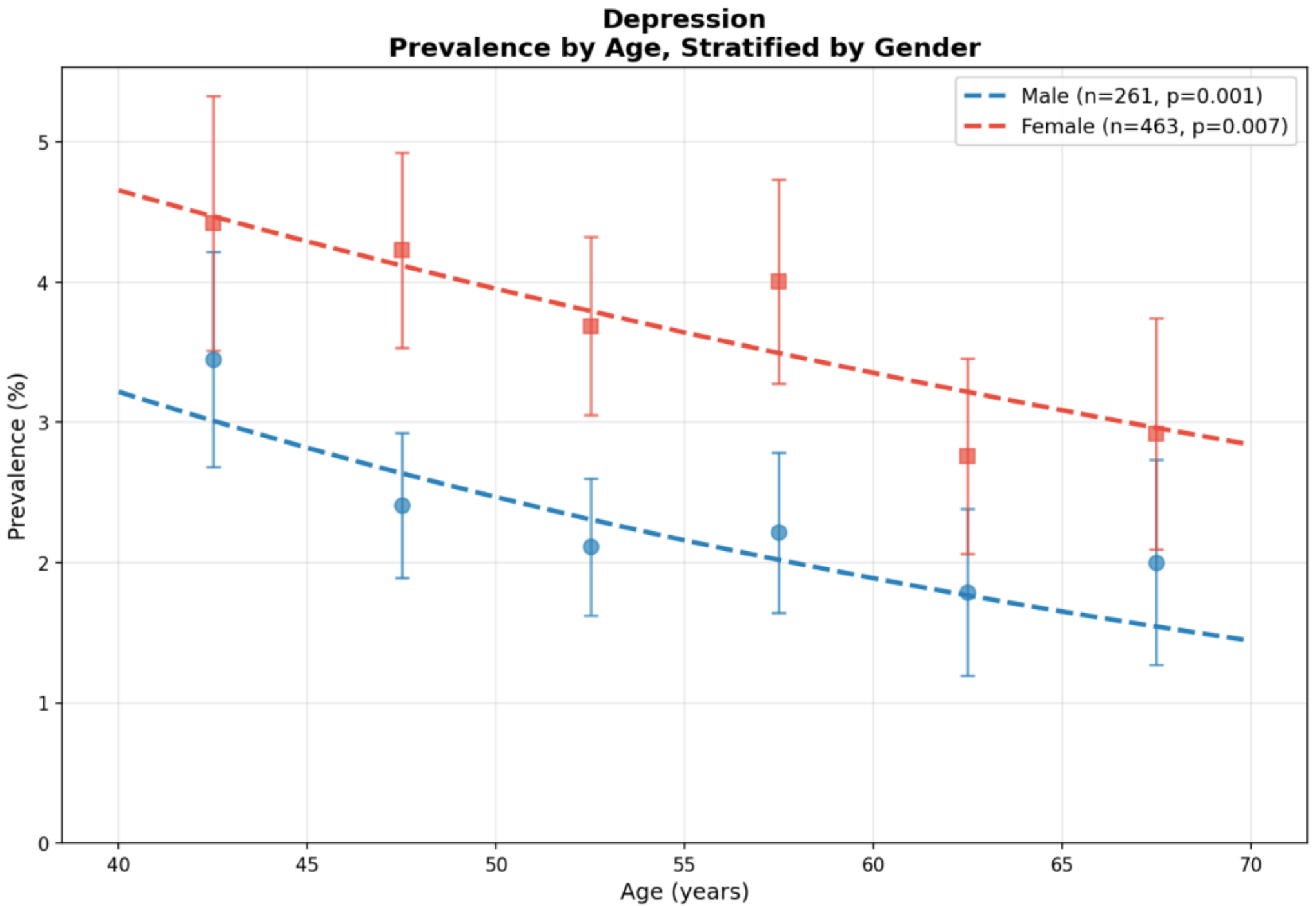
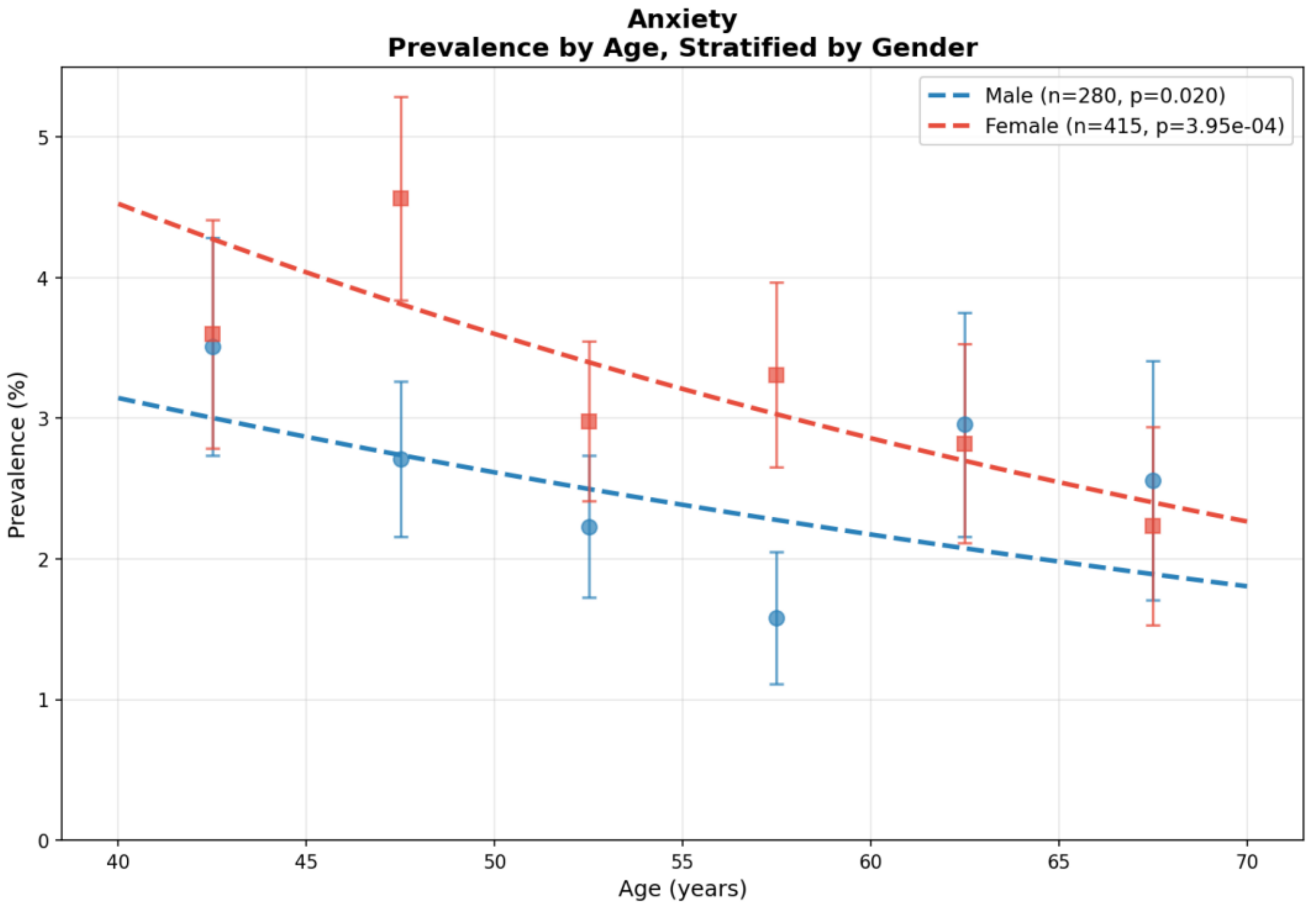
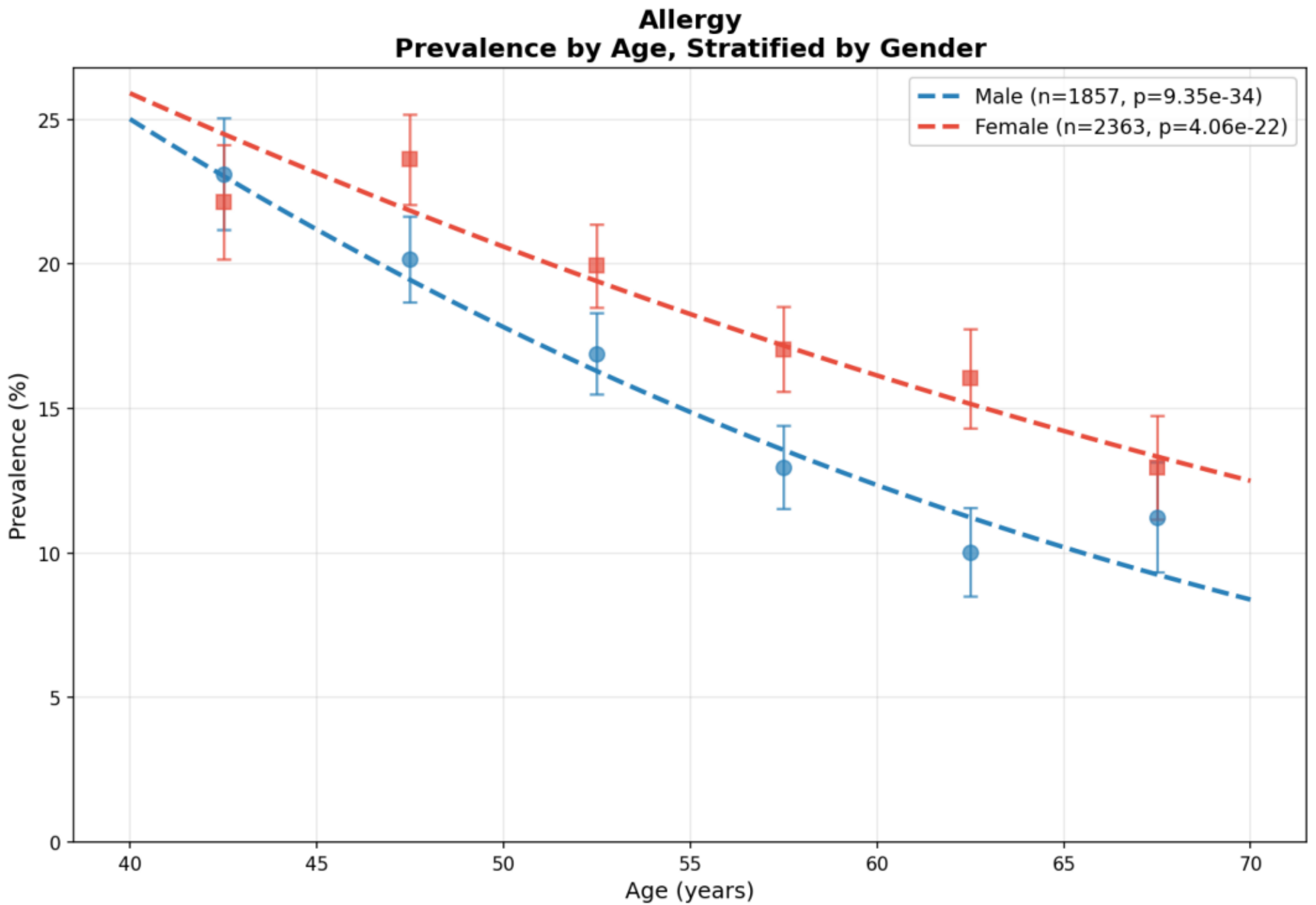
6
7
40
45
50
55
60
65
70


Osteoarthritis
Prevalence by Age, Stratified by Gender
Male (n=216, p=4.34e-15)
Female (n=395, p=1.83e-50)
Prevalence (%)
Age (years)
0
2
4
6
8
10
12
40
45
50
55
60
65
70


Depression
Prevalence by Age, Stratified by Gender
Male (n=261, p=0.001)
Female (n=463, p=0.007)
Prevalence (%)
Age (years)
0
1
2
3
4
5
40
45
50
55
60
65
70


Anxiety
Prevalence by Age, Stratified by Gender
Male (n=280, p=0.020)
Female (n=415, p=3.95e-04)
Prevalence (%)
Age (years)
0
1
2
3
4
5
40
45
50
55
60
65
70


Allergy
Prevalence by Age, Stratified by Gender
Male (n=1857, p=9.35e-34)
Female (n=2363, p=4.06e-22)
Prevalence (%)
Age (years)
0
5
10
15
20
25
40
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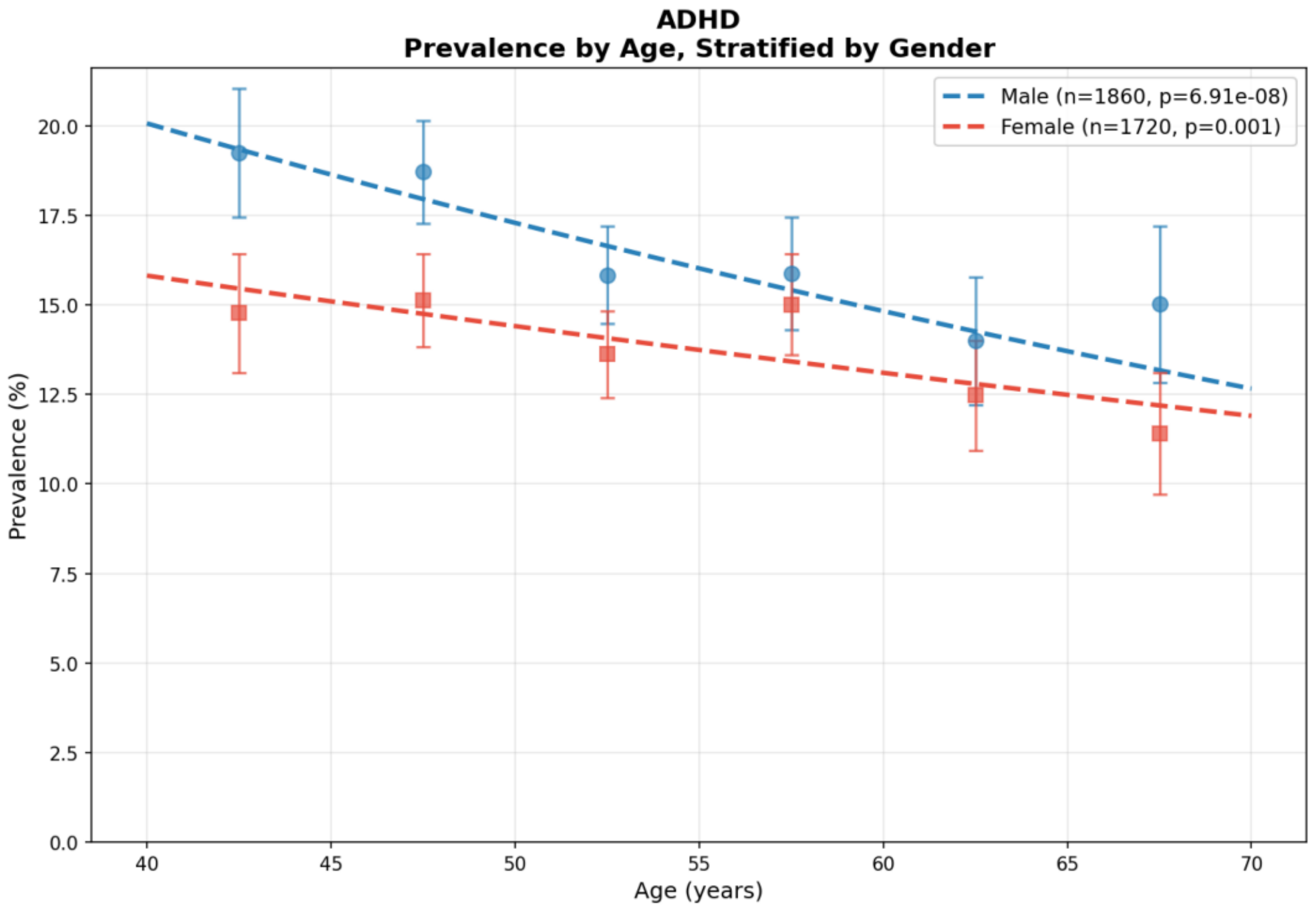
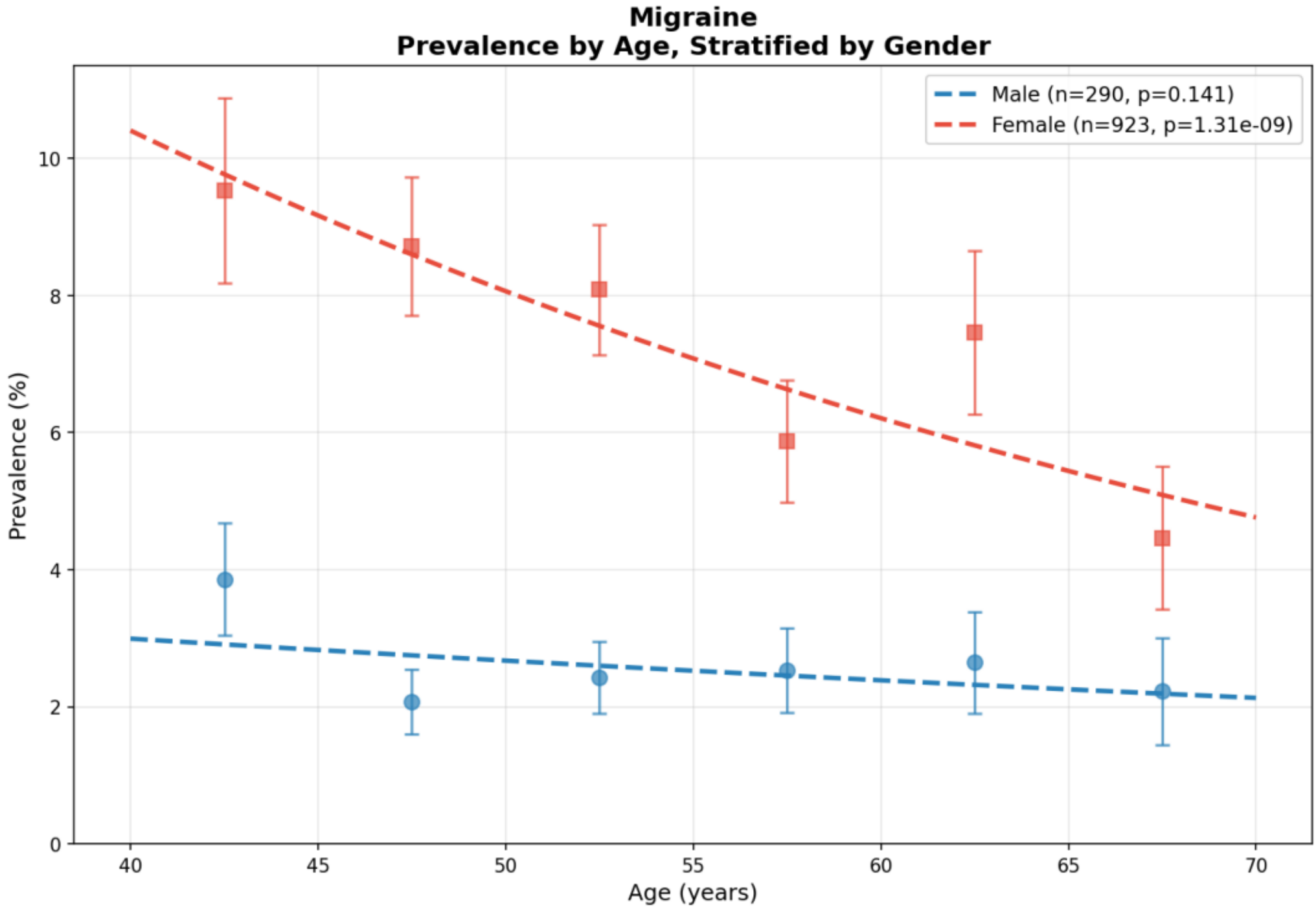
55
60
65
70


ADHD
Prevalence by Age, Stratified by Gender
Male (n=1860, p=6.91e-08)
Female (n=1720, p=0.001)
Prevalence (%)
Age (years)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
40
45
50
55
60
65
70


Migraine
Prevalence by Age, Stratified by Gender
Male (n=290, p=0.141)
Female (n=923, p=1.31e-09)
Prevalence (%)
Age (years)
0
2
4
6
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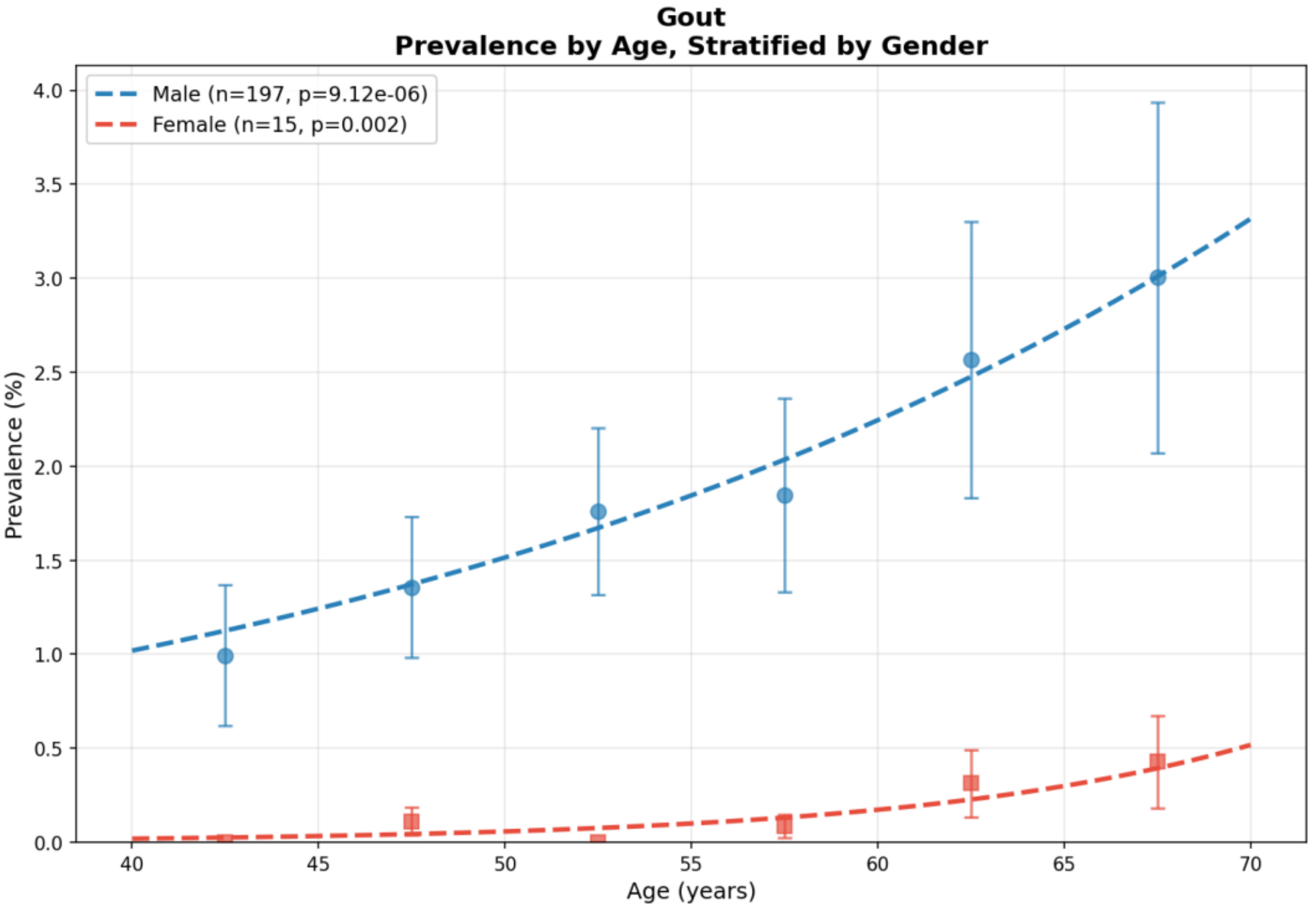
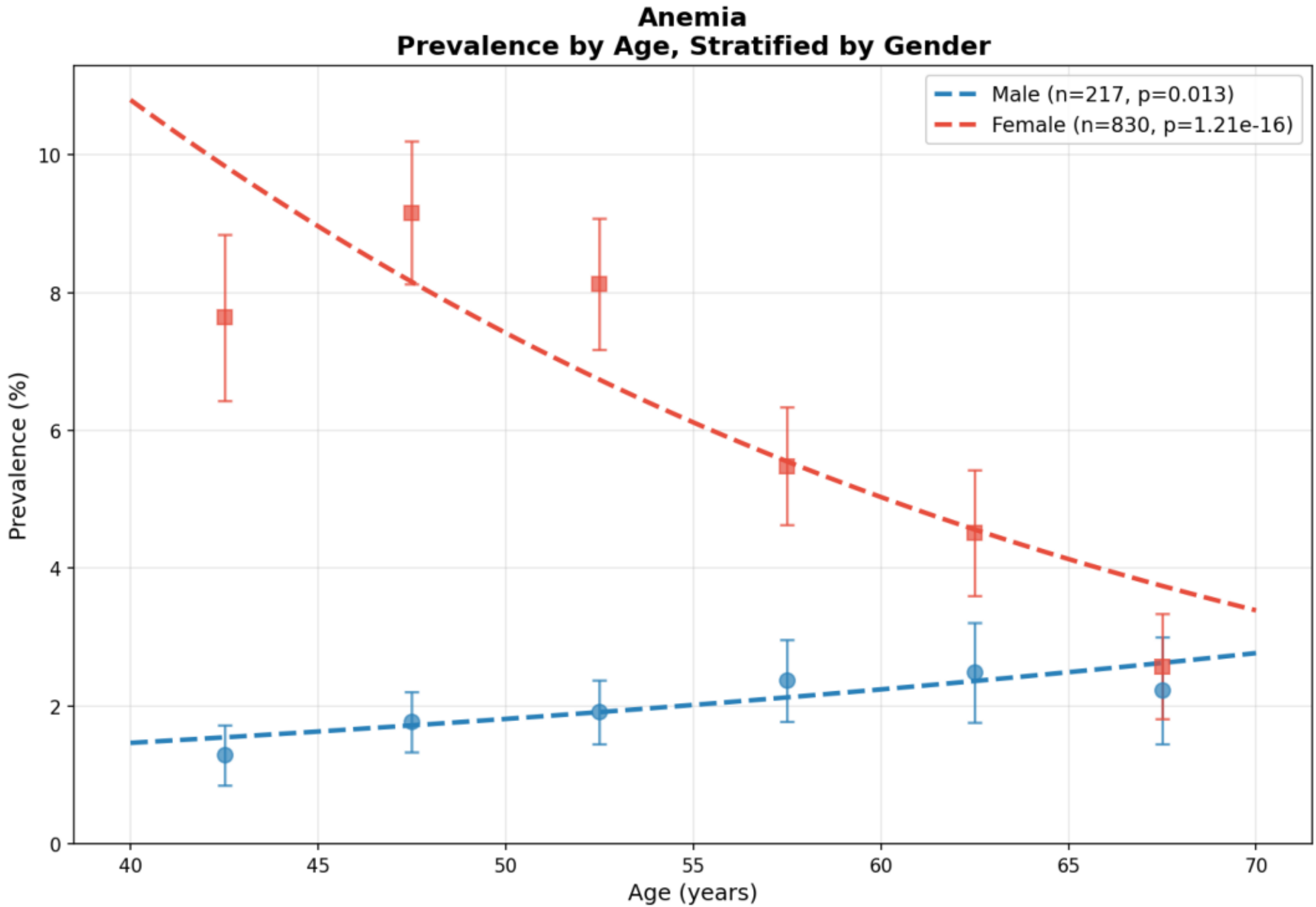

Gout
Prevalence by Age, Stratified by Gender
Male (n=197, p=9.12e-06)
Female (n=15, p=0.002)
Prevalence (%)
Age (years)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
40
45
50
55
60
65
70


Anemia
Prevalence by Age, Stratified by Gender
Male (n=217, p=0.013)
Female (n=830, p=1.21e-16)
Prevalence (%)
Age (years)
0
2
4
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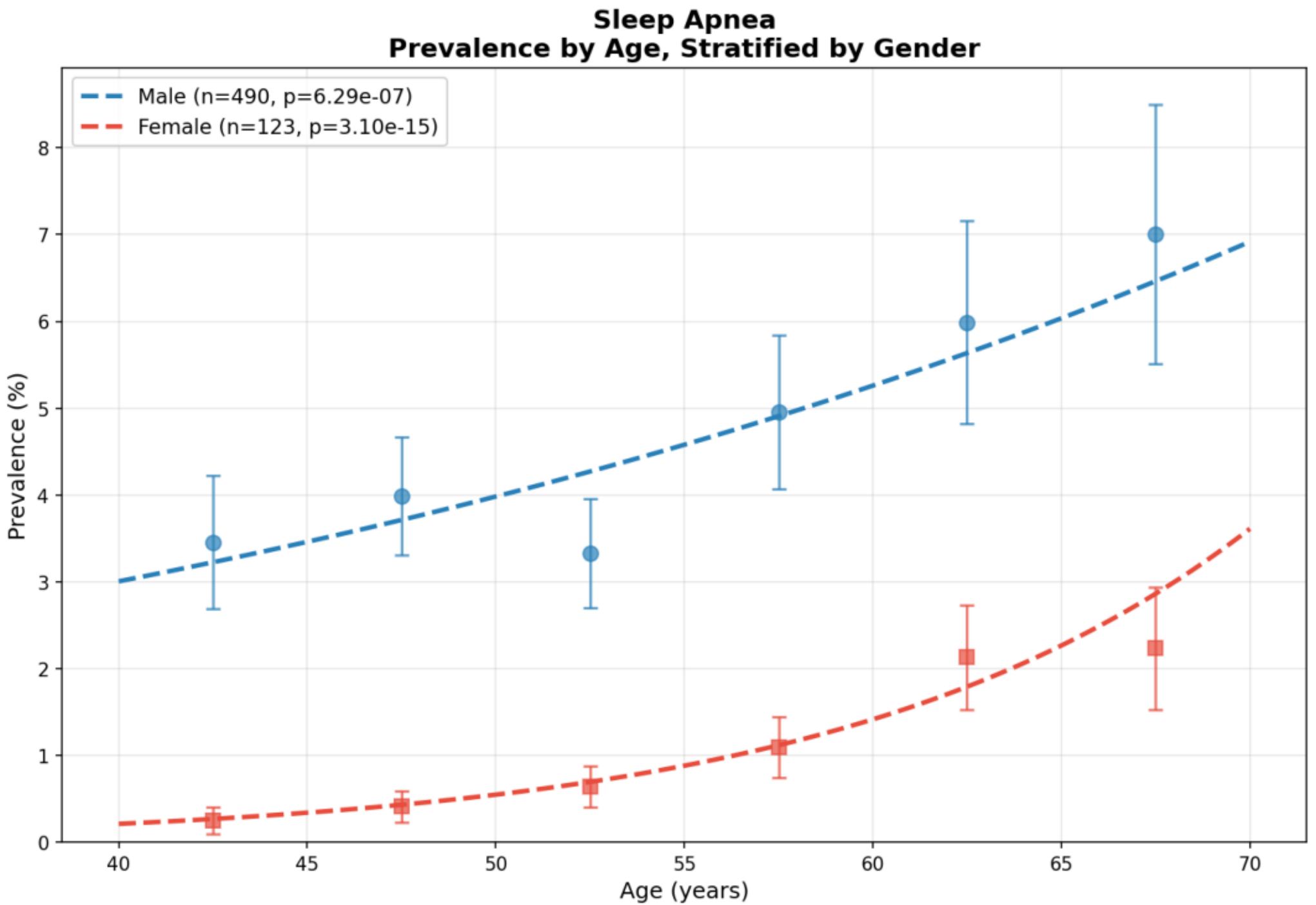
Sleep Apnea
Prevalence by Age, Stratified by Gender
Male (n=490, p=6.29e-07)
Female (n=123, p=3.10e-15)
Prevalence (%)
Age (years)
0
1
2
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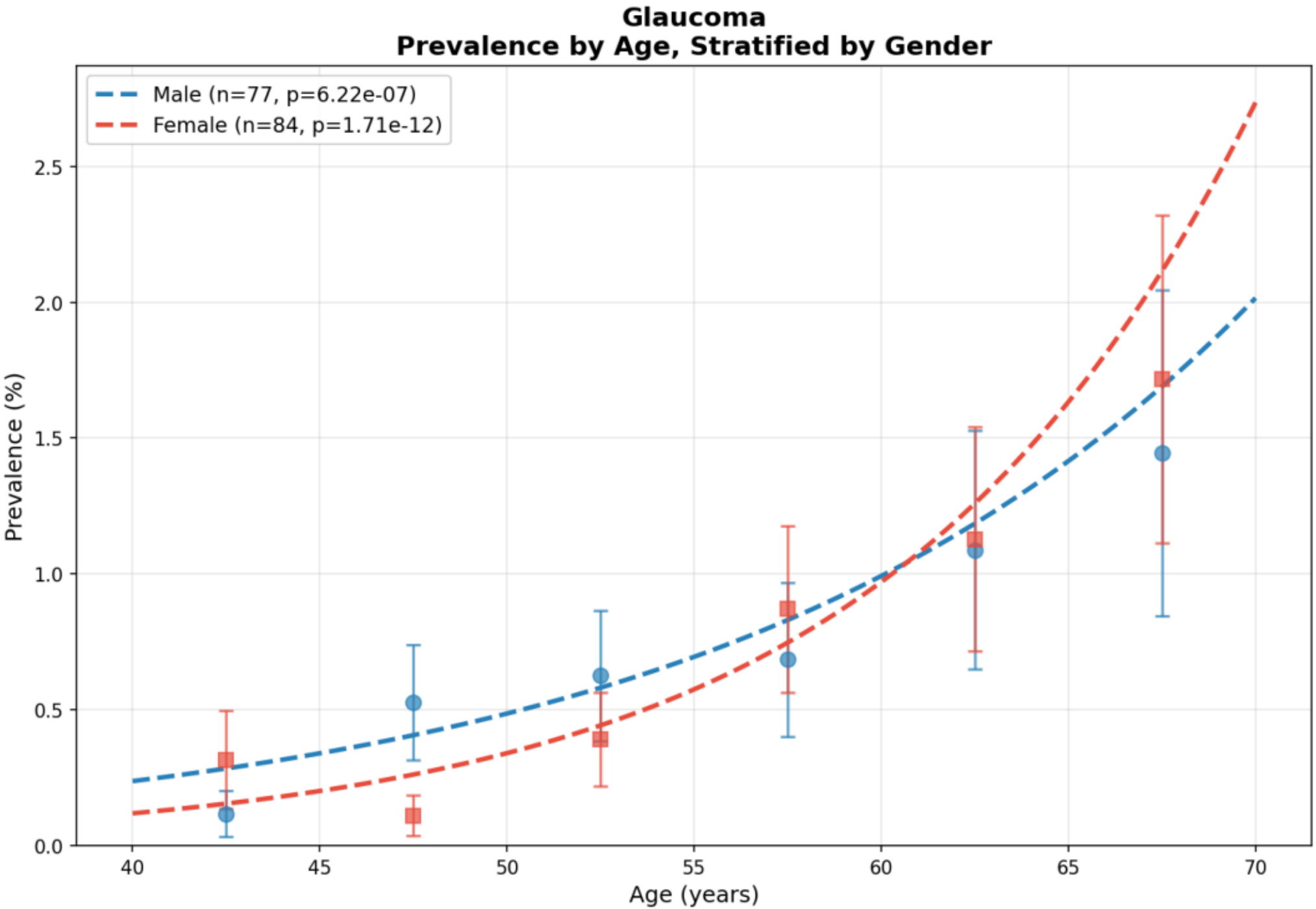
70


Glaucoma
Prevalence by Age, Stratified by Gender
Male (n=77, p=6.22e-07)
Female (n=84, p=1.71e-12)
Prevalence (%)
0.0
0.5
1.0
1.5
2.0
2.5
40
45
50
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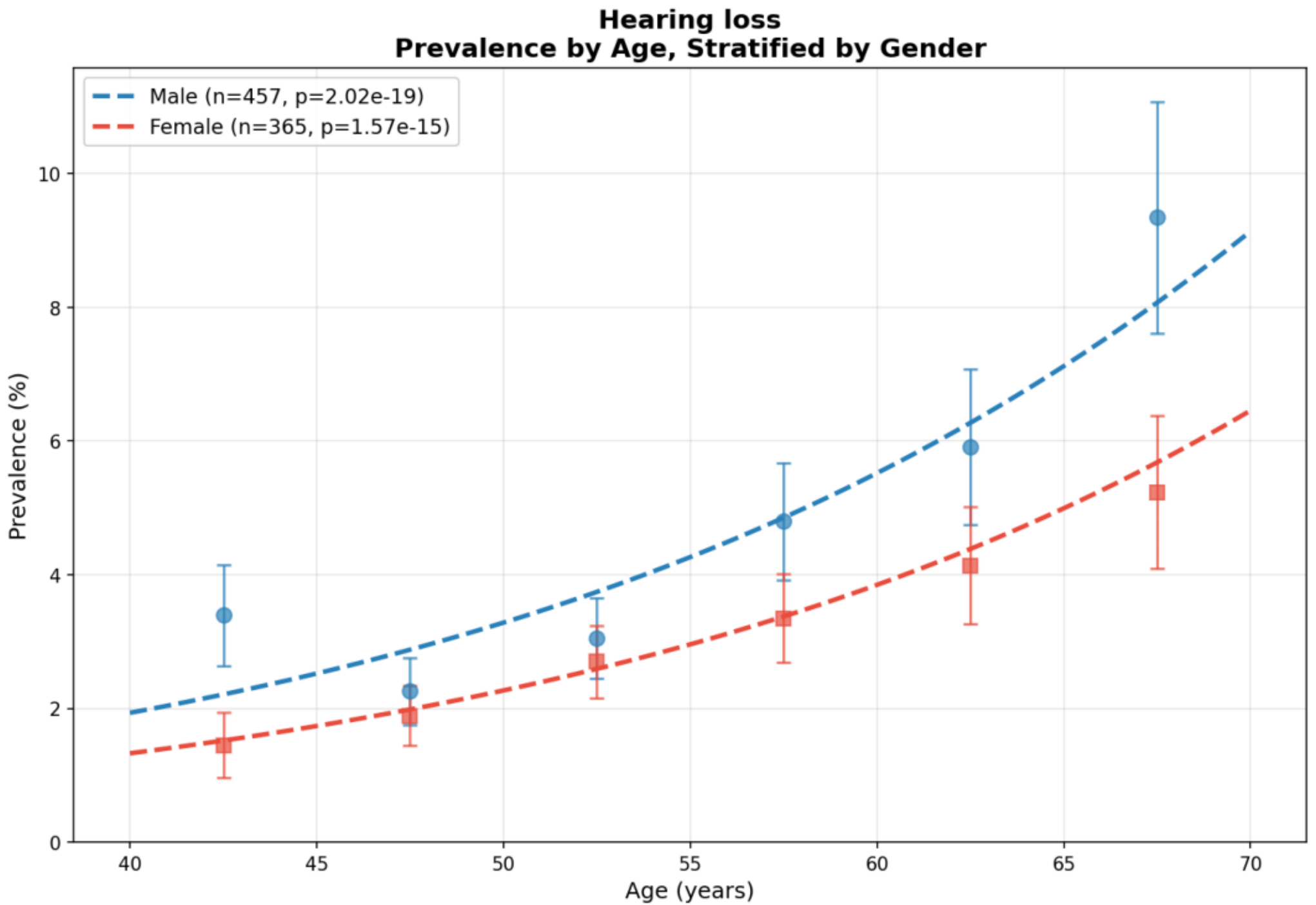
60
65
70
Age (years)


Hearing loss
Prevalence by Age, Stratified by Gender
Male (n=457, p=2.02e-19)
Female (n=365, p=1.57e-15)
Prevalence (%)
Age (years)
0
2
4
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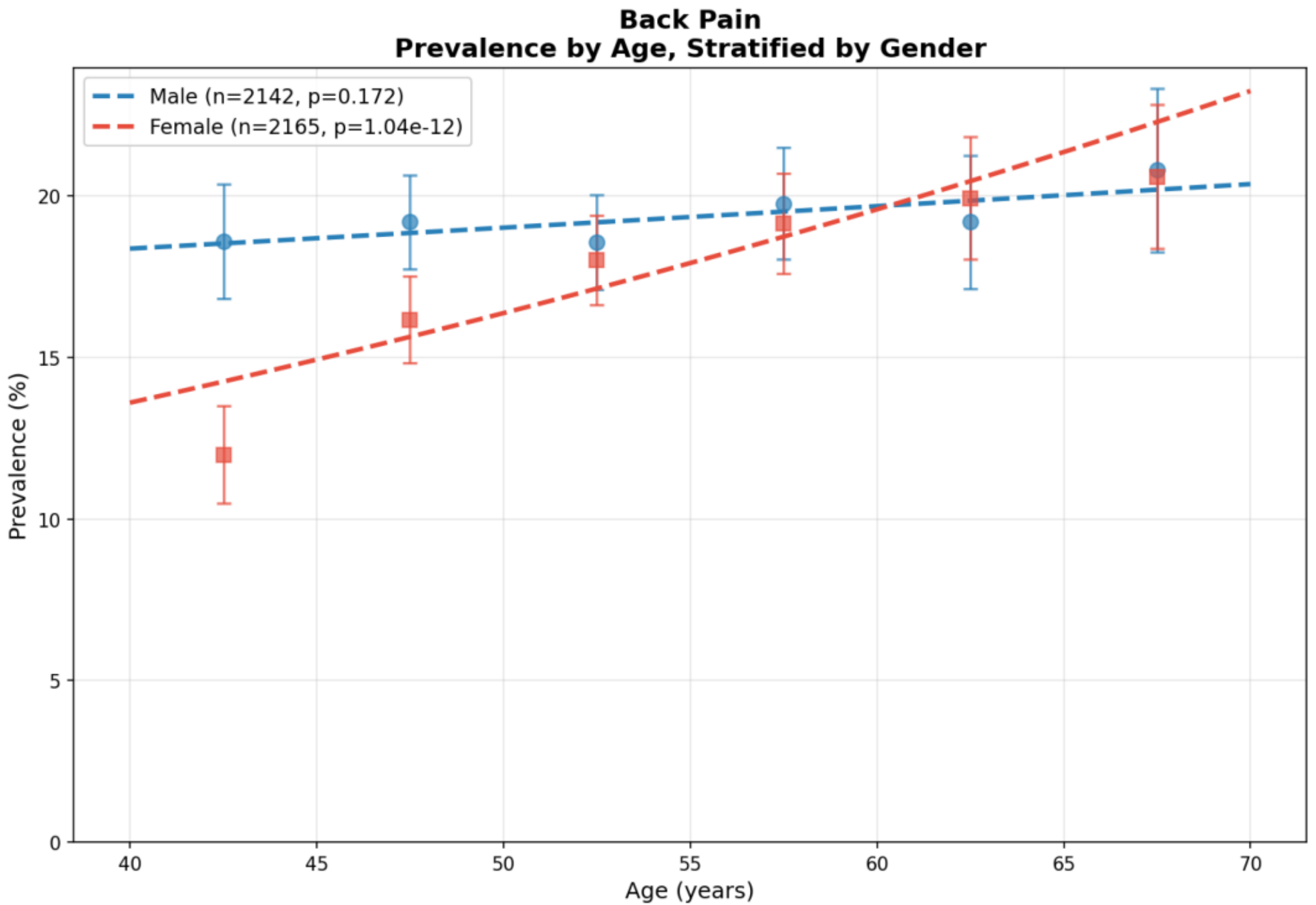
Back Pain
Prevalence by Age, Stratified by Gender
Male (n=2142, p=0.172)
Female (n=2165, p=1.04e-12)
Prevalence (%)
0
5
10
15
20
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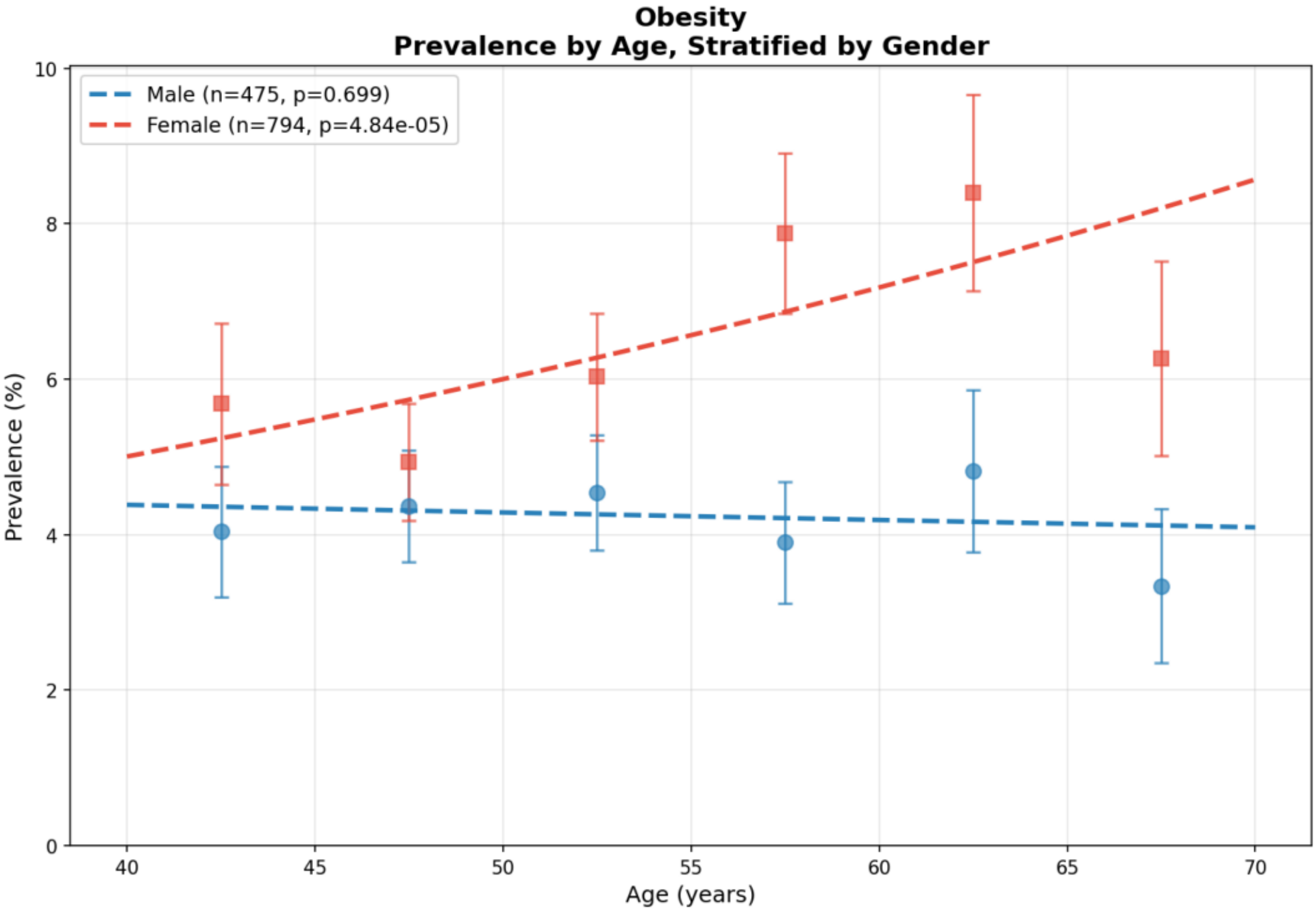
60
65
70
Age (years)


Obesity
Prevalence by Age, Stratified by Gender
Male (n=475, p=0.699)
Female (n=794, p=4.84e-05)
Prevalence (%)
Age (years)
0
2
4
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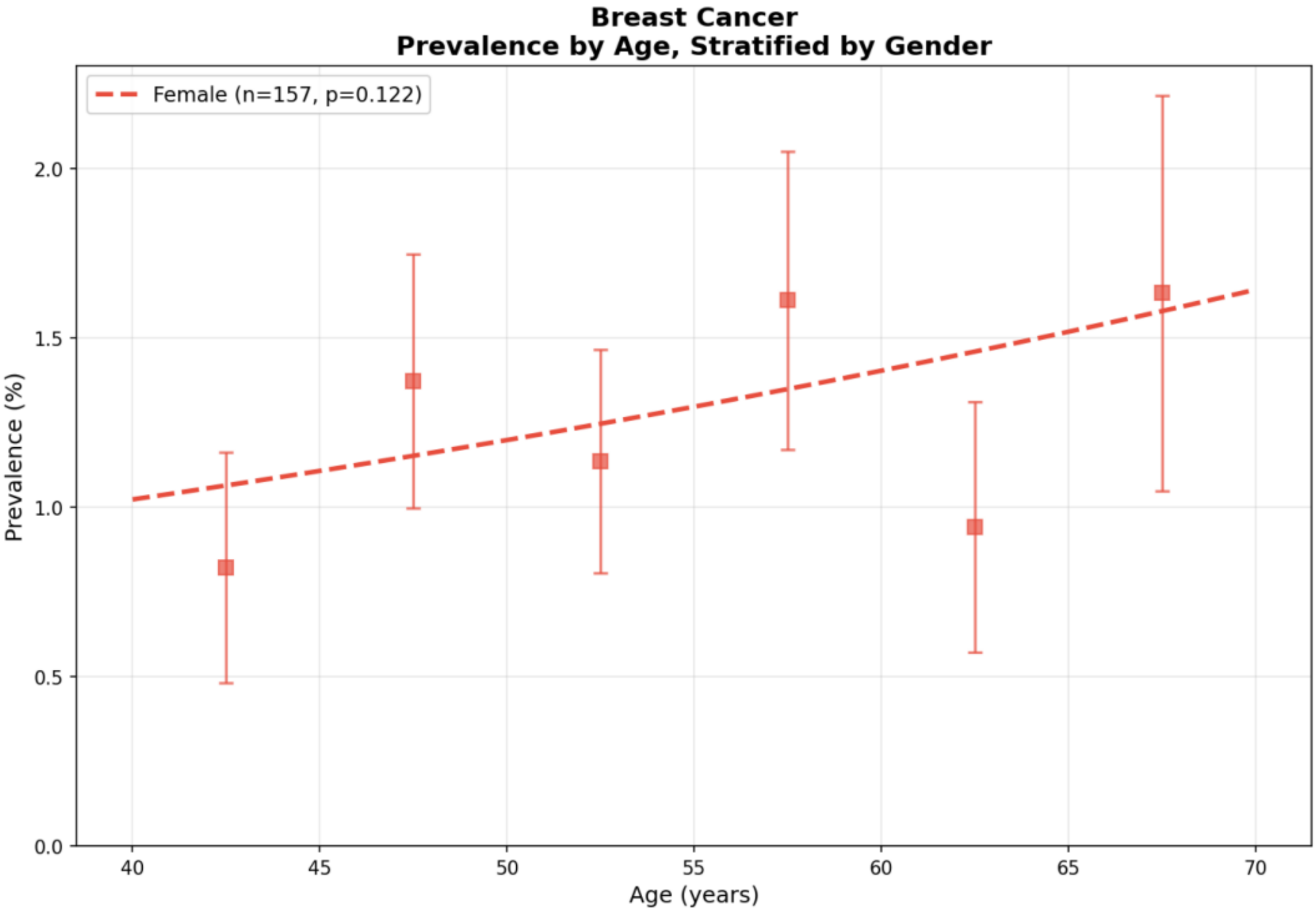
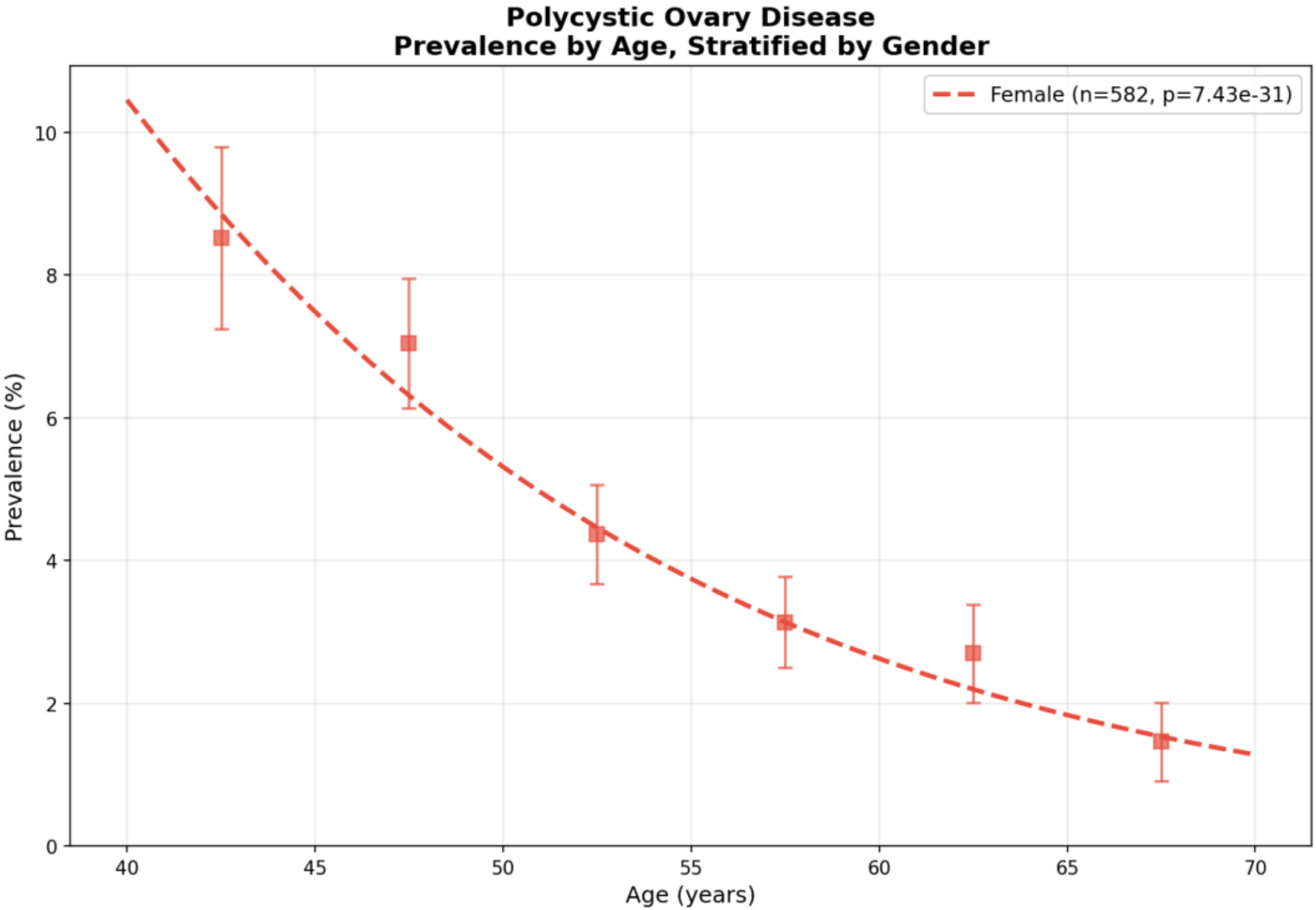
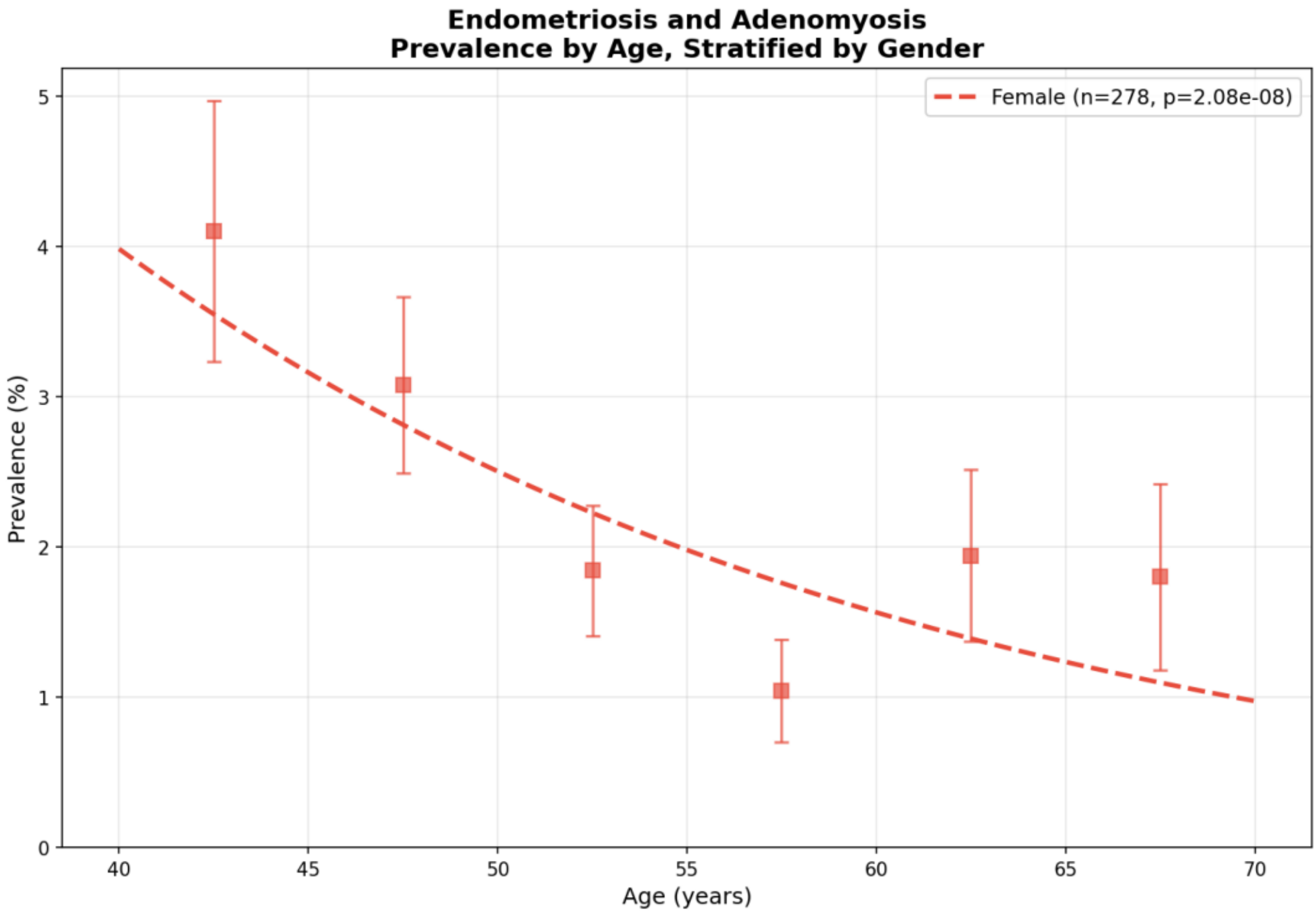
Breast Cancer (Sex-Specific)



Polycystic Ovary Disease (Sex-Specific)



Endometriosis and Adenomyosis (Sex-Specific)



METHODOLOGY

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STATISTICAL MODEL

- Separate logistic regressions for each gender
- $\log(p/(1-p)) = \beta_0 + \beta_1 \times \text{Age}$
- Performed independently for males and females
- Prevalence shown as actual percentage (%)

PLOT INTERPRETATION

- Blue circles/line = Male prevalence and fitted curve
- Red squares/line = Female prevalence and fitted curve
- Error bars = 95% confidence intervals (Wilson score)
- Dashed lines = Fitted logistic regression curve
- Y-axis = Prevalence (%) - actual percentage with condition
- Legend shows sample size (n) and p-value

ODDS RATIO INTERPRETATION

- OR/Decade = exp(coefficient × 10)
- OR = 2.0 means 2x more likely per decade
- OR = 0.5 means 2x less likely per decade

INCLUSION CRITERIA

- Age: 40-70 years
- Minimum 50 cases per gender for analysis
- Total n ≥ 100 for condition

LIMITATIONS

1. Some conditions are sex-specific (PCOS, breast cancer)
2. Prevalence differences may affect power
3. Hormonal factors not directly measured
4. Cross-sectional design limitations apply

SAMPLE SIZES

- Total: 23,473 subjects
- Female: 12,313 (52%)
- Male: 11,152 (48%)

KEY FINDINGS BY CONDITION CATEGORY
{'='*55}

CARDIOVASCULAR
Both genders show increased risk with age for:
• Hypertension, Ischemic Heart Disease, Heart Valve Disease
Males may have steeper increase for some conditions.

METABOLIC
• Diabetes, Hyperlipidemia: Increase with age in both
• Obesity: Different patterns by gender
• Fatty Liver: More common in males

MENTAL HEALTH
• Depression: Decreases with age in both genders
• Anxiety: Similar pattern, may differ in magnitude
• ADHD: Cohort effect visible in both genders

MUSCULOSKELETAL
• Osteoarthritis: Increases with age, similar in both
• Gout: Much more common in males, steeper increase

AUTOIMMUNE/INFLAMMATORY
• Allergy: Decreases with age in both
• Psoriasis: Different patterns by gender possible
• Hashimoto's: More common in females

SENSORY
• Hearing loss: Increases in both, may differ in rate
• Glaucoma: Similar pattern in both genders

NOTES ON INTERPRETATION
• Gender differences may reflect:
  - Biological differences (hormones, genetics)
  - Healthcare-seeking behavior
  - Diagnostic patterns
  - Survival differences
• Some apparent differences may not be statistically significant
• Always check sample sizes and p-values